

Progress Report: Milestone 2 (Initial EDA)

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For the initial EDA for the Vibe Finder project, I loaded the dataset (TidyTuesday Spotify audio features, 32,833 tracks), documented it, and checked it for data quality problems.

What I've Done So Far

- Loaded the dataset and checked that it loads correctly from the source URL given in the client's proposal.
- Wrote a column dictionary for the 9 audio features I'm planning to use for clustering (danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo).
- Checked for data quality issues and found about 13.6% duplicate track_id rows, 5 rows with missing metadata, and 6 rows where loudness is above 0 dB, which doesn't really make sense for that feature.
- Looked at the distribution of each audio feature. A few of them (speechiness, acousticness, instrumentalness, liveness) are heavily right-skewed, and loudness/tempo are on a completely different scale than the 0-1 features, so I know I'll need to standardize everything before clustering.
- Checked correlations between the features (e.g. energy and loudness are positively correlated) just to understand if any features are redundant.
- Looked at the existing playlist_genre distribution, but only as a reference. It won't be used to build the clusters, just to check later whether the clusters end up mixing genres, since that's part of the client's success criteria.
- Asked the client (Kaleb Vinthers) some clarifying questions on Discord about which audio features to use and how strict the target cluster count (5-10) is. He said both were flexible, so I logged that in the notebook's Assumptions Log.

Next Plan (Checkpoint 2)

- Clean the dataset (remove duplicates, drop the rows with missing data) based on what I found above.
- Standardize the 9 features and run K-means with $k = 5-10$, and pick the best k using the elbow method / silhouette score.
- Interpret the clusters and check if they actually mix genres like the client wants.